

Data-Driven Testing with Apache POI

Implementing Scalable Parametrization in Selenium WebDriver



```
CodeEditor.java

public void testData() {
    // 
    String data = ExcelUtil.read("data.xlsx", 1, 1);
    // 
    driver.findElement(By.id("input")).sendKeys(data);
}
```

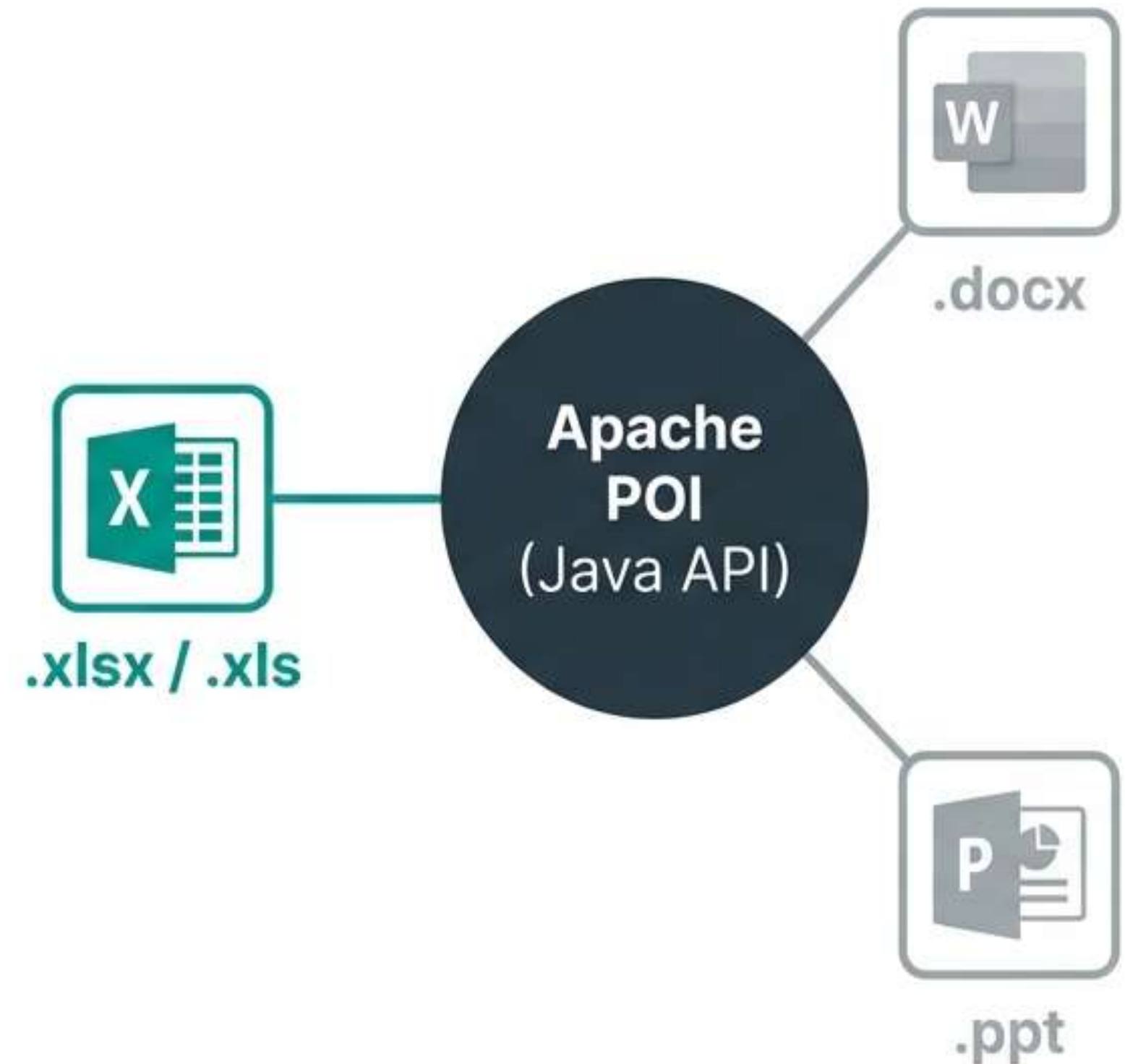
	A	B	C	D
	Test Case ID	Username	Password	Expected Result
	TC001	user1	pass1	Success
	TC002	user1	pass2	Failure
	TC003	user2	pass3	Success
	TC004	user4	pass4	Success

What is Apache POI?

Definition: A robust Java API enabling programmers to modify Microsoft Office documents.

Role: Middleware between Selenium WebDriver tests and Excel data files.

Key Capability: Extract massive datasets to validate business logic without altering code.



Project Configuration & Dependencies

Add the following to your `pom.xml` file to enable Excel interaction.

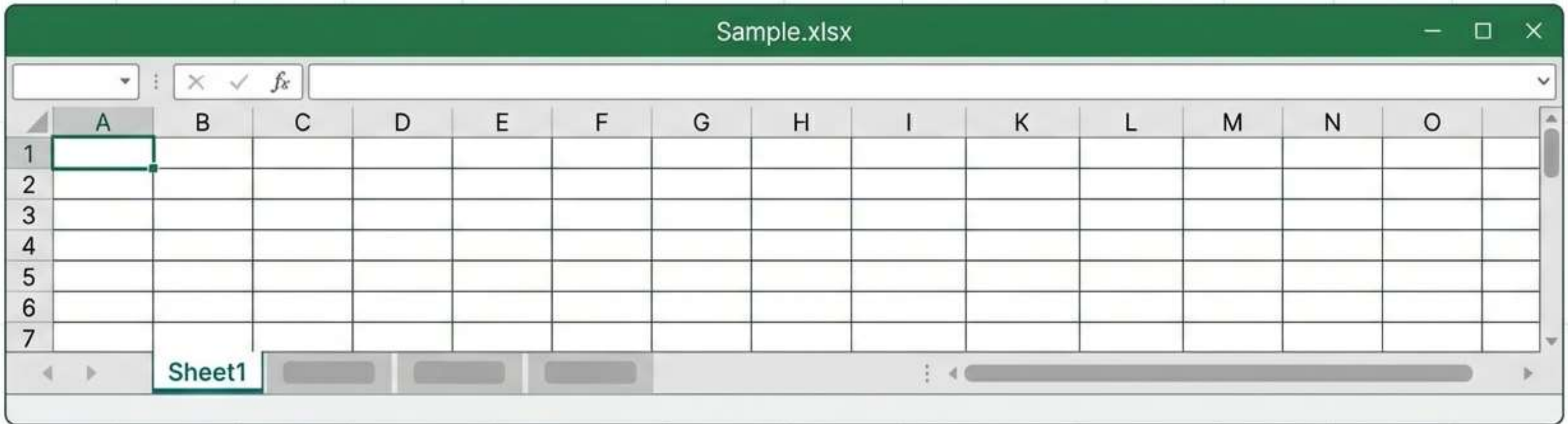
```
<!-- Core Library -->
<dependency>
  <groupId>org.apache.poi</groupId>
  <groupId>org.apache.poi</groupId>
  <artifactId>poi</artifactId>
  <version>5.2.3</version>
</dependency>

<!-- OOXML for .xlsx files -->
<dependency>
  <groupId>org.apache.poi</groupId>
  <artifactId>poi-ooxml</artifactId>
  <version>5.2.3</version>
</dependency>
```

Note

⚠️ Ensure versions (5.2.3) match exactly to avoid runtime conflicts.

Implementation: Accessing the Sheet



```
// Step 1: Instantiate the Workbook  
XSSFWorkbook wbook = new XSSFWorkbook("./Data/Sample.xlsx");
```

```
// Step 2: Access the specific sheet  
// (Index 0 accesses the first tab)
```

```
XSSFSheet sheet = wbook.getSheetAt(0);
```

Zero-based indexing

Implementation: Extracting Cell Data

Row Index: 0

	A	B	C	D
1				
0		John Doe	Mark	Martin

Column Index: 1

```
// Step 3 & 4: Accessing the Row and Cell, Step 5: Reading Data
// Step 3: Access the specific row (Row 0 = First Row)
XSSFRow row = sheet.getRow(0);

// Step 4: Access the specific cell (Column 1 = Second Column)
XSSFCell cell = row.getCell(1);

// Step 5: Read the string value
String data = cell.getStringCellValue();
```


Advanced Challenges: Data Types



The Problem

Excel stores IDs and Phone Numbers as **Numeric**. Reading them with `'getStringCellValue()'` throws an `'IllegalStateException'`.

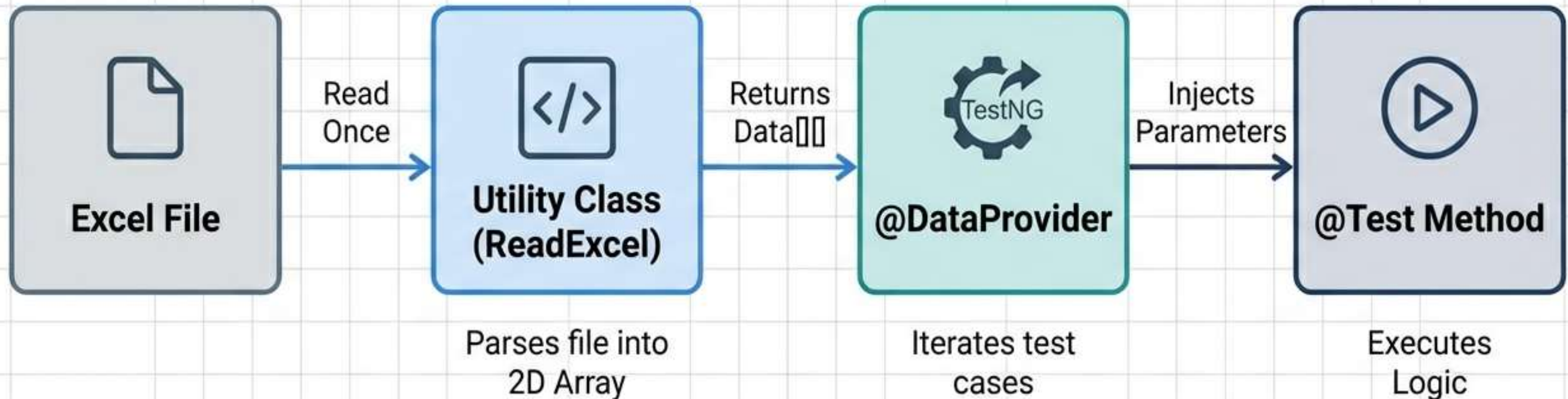


The Solution

Check the type or force a conversion before reading.

```
// Force the cell to be treated as String  
cell.setCellType(CellType.STRING);  
String data = cell.getStringCellValue();
```

Advanced Challenges: Performance &



Avoid opening the file inside the Test method. Read once, run many.

Key Takeaways



1. **Hierarchy:**



2. **Dependencies:** Both `poi` and `poi-ooxml` are mandatory.



3. **Method:** `getStringCellValue()` is the standard for text.



4. **Indexing:** Apache POI uses **0-based indexing**.



5. **Stability:** Always close resources (`wbook.close()`).